How Does Literacy Rate Affect the Prevalence of HIV?

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Abstract

This paper examines the relationship between the literacy rate of the population aged 15–24 and the prevalence of Human Immunodeficiency Virus (HIV) among individuals aged 15–49, using country-level panel data from 2000 to 2021. HIV prevalence is recoded as a binary indicator equal to one when a country-year exceeds the worldwide median of 0.4 percent. We estimate a linear probability model, a quadratic specification, a probit model, and a fixed effects model that controls for life expectancy and for country and year effects. Literacy is negatively and significantly associated with high HIV prevalence in the three models without fixed effects, but the magnitude of the association is economically small in every specification, and the coefficient is not statistically significant once country and year fixed effects, and life expectancy are included. We discuss omitted variable bias, sample selection, simultaneous causality, and measurement error as threats to internal validity, and we assess the external validity of the results.

1. Introduction

This paper explores the relationship between the percentage of people aged 15–49 who are infected with HIV (Y) and the percentage of people aged 15–24 who can both read and write, with understanding, a short, simple statement about their everyday life (X_1). The World Bank notes that a literate individual "can seek and use information for the betterment of the health, nutrition and education of their household members" (World Bank, n.d.-b).

We hypothesize that in countries with less access to education, and therefore lower literacy rates, individuals have fewer opportunities to advance their position in society and may turn to alternative sources of income, such as sex work. Ngugi et al. (n.d.) report that East and Southern Africa remain the regions most affected by HIV, with 34 percent of the global burden of infection concentrated in ten Southern African countries. This concentration is largely attributable to the prevalence of young female sex workers and of a clientele that lacks the

resources and education to practice safe sex, seek medical treatment, and undergo regular testing for sexually transmitted infections (STIs).

Testing and awareness are also limited more broadly. In a 2009 progress report on HIV in low- and middle-income countries, the World Health Organization (n.d.) found that 56 percent of HIV-positive respondents did not know their status, either because they had not been tested or because they had not received their results. Of the remaining 44 percent who believed they knew their status, 28 percent mistakenly believed they were HIV-negative, and only 16 percent correctly knew they were HIV-positive. UNAIDS further estimates that fewer than 50 percent of sex workers worldwide have access to HIV prevention programs (Ngugi et al., n.d.).

Understanding the effect of literacy on HIV prevalence can help policymakers design educational outreach and health care initiatives that target vulnerable populations. Educational programs, particularly those directed toward populations with low literacy, could substantially reduce HIV transmission and allow individuals to regain agency over their health.

2. Data

2.1 HIV prevalence (dependent variable)

Our dependent variable (Y) is the prevalence of HIV, total (percent of population aged 15–49) (World Bank, n.d.-c), which measures the percentage of people aged 15–49 who are infected with HIV. In our code, the continuous measure is named HIV and the binary measure is named highHIV. The data are UNAIDS estimates, aggregated as a weighted average, and are based on survey and surveillance data that UNAIDS continually updates to track trends in HIV/AIDS. UNAIDS (2023) draws on sources including antenatal clinics, surveys of key populations or of the entire population, antiretroviral therapy use, and, more recently, HIV case reporting and vital registration systems. The underlying models account for changes in urbanization and for medical advances; improvements in treatment have lengthened the life expectancy of infected individuals.

To clean the data, we dropped the variables unnecessary for our analysis (country code, indicator code, and indicator name), which are identifiers rather than numerical data. We then renamed the remaining variables, originally labeled v5 through v66, using a loop so that each corresponds to its year of collection (1990–2021). To make each observation a country-year

rather than a country, we reshaped the data from wide to long format. Finally, we verified that the number of non-missing observations exceeded 100; the HIV variable has 3,079.

Because we estimate a probit model, we recoded HIV as a binary variable. The worldwide median of HIV is 0.4 percent, so we generated highHIV, equal to one if HIV exceeds 0.4 percent and zero otherwise. A value of one therefore indicates that a country's HIV prevalence among individuals aged 15–49 is above the worldwide median. The variable ranges from 0 to 1, with a sample mean of 0.5001.

2.2 Literacy rate (independent variable of interest)

Our independent variable of interest (X_1) is the youth literacy rate, total (percent of people aged 15–24) (World Bank, n.d.-b), which measures the percentage of young adults who can "both read and write with understanding a short simple statement about their everyday life" (variable name in code: LIT). The data are collected by the UNESCO Institute for Statistics (UIS) through national censuses and household surveys, supplemented by the Global Age-Specific Literacy Projection Model (GALP) for countries without recent literacy data. We focus on ages 15–24 because this group has completed schooling more recently, so the measure reflects recent progress in education better than a measure covering the entire population. The data were cleaned in the same way as the HIV data. The variable contains 950 observations, ranges from 13.14 to 100 percent, and has a sample mean of 84.40 percent.

2.3 Life expectancy (control variable)

Our second independent variable (X_2), included as a control in the fourth regression, is life expectancy at birth, total (years) (World Bank, n.d.-a), which represents the number of years a newborn would be expected to live given prevailing mortality patterns (variable name in code: LIFE). We include this control because life expectancy is an important indicator of population health and a proxy for the accessibility of health care. The data come from the United Nations Population Division's *World Population Prospects*, an annual series calculated from period life tables that capture the mortality experience of a population in a given year. Unlike our HIV and literacy variables, which are restricted to specific age groups, life expectancy describes the entire population. The data were cleaned in the same way as the HIV and literacy data. The variable contains 15,866 observations, ranges from 11.99 to 85.49 years, and has a sample mean of 64.25 years.

2.4 Final dataset

For the final dataset, we dropped all years before 2000, so the analysis examines the relationship between HIV prevalence (highHIV) and the literacy rate (LIT) from 2000 to 2021. Because HIV prevalence is measured for ages 15–49 and literacy for ages 15–24, the data are intended to describe the youth and young-adult population. For the HIV and literacy variables jointly, there are 751 non-missing observations. The final dataset includes numerous countries of differing income levels and geography, so, in principle, the results can be applied worldwide. Figure 1 presents a scatter plot of the final data.

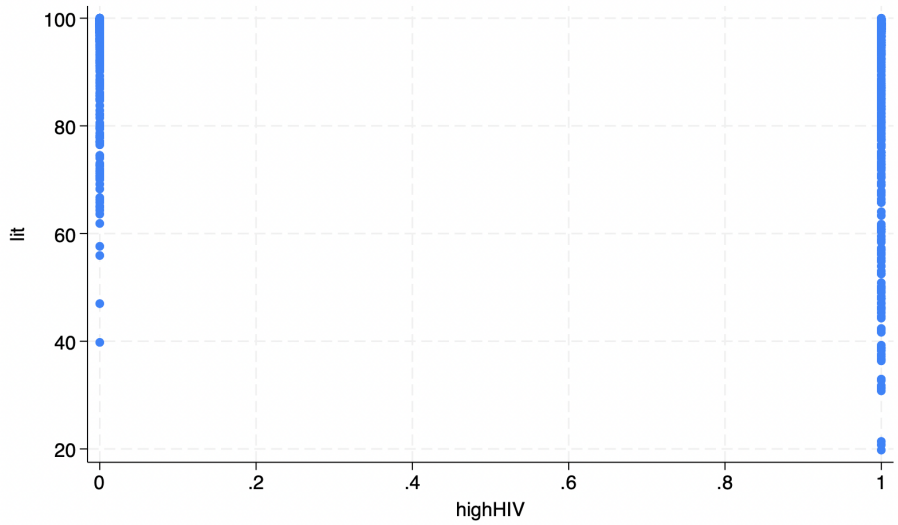


Figure 1. Youth literacy rate (lit) by high HIV prevalence indicator (highHIV).

3. Empirical Specification and Results

3.1 Model 1: Linear probability model

Our first, naive model regresses highHIV on the literacy rate with no controls. It is estimated by ordinary least squares (OLS) with robust standard errors:

$$\text{highHIV} = 1.321 - 0.009\ (\text{literacy rate})$$

The fitted line is shown in Figure 2. The coefficient on the literacy rate is −0.009: a one-percentage-point increase in a population's literacy rate decreases by 0.009 the probability that the population is located in an area where HIV prevalence exceeds 0.4 percent. The coefficient is statistically significant at the 10, 5, and 1 percent levels, providing evidence that the literacy rate of a population aged 15–24 is associated with HIV prevalence in the population aged 15–49. The

coefficient is not economically significant. Relative to the median HIV prevalence of 0.4 percent as a reference point, (0.009 / 0.4) × 100 = 2.25 percent, which we consider economically small.

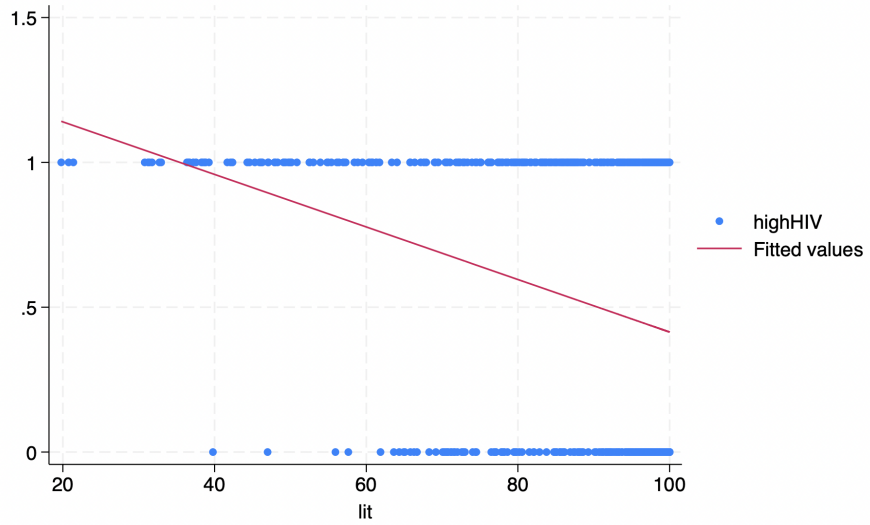


Figure 2. Linear probability model: fitted values of highHIV against the literacy rate.

3.2 Model 2: Quadratic specification

The second model is a polynomial (quadratic) regression, also estimated by OLS with robust standard errors:

$$\text{highHIV} = 1.084 - 0.0019\ (\text{literacy rate}) - 0.00004\ (\text{literacy rate}^2)$$

Because the specification is quadratic, the coefficients on X_1 and X_1^2 cannot be interpreted directly. The coefficient on the literacy rate is −0.0090703, and the coefficient on its square is −0.0000486; neither is statistically significant at the 10, 5, or 1 percent level. To examine how X_1 relates to Y, we therefore analyze the fitted quadratic curve, shown in Figure 3. The curve indicates that as the literacy rate rises, the probability of high HIV prevalence falls, with diminishing returns: the marginal effect of an additional percentage point of literacy becomes smaller as literacy increases, so the slope of the fitted line becomes less steep. In other words, a more literate population is associated with lower HIV prevalence. The curvature of the fitted line is not substantial, which indicates that the quadratic term is not economically significant. For the linear term, we assess economic significance as in Model 1: (0.0019 / 0.4) × 100 = 0.475 percent relative to the median HIV prevalence, which we likewise consider economically small.

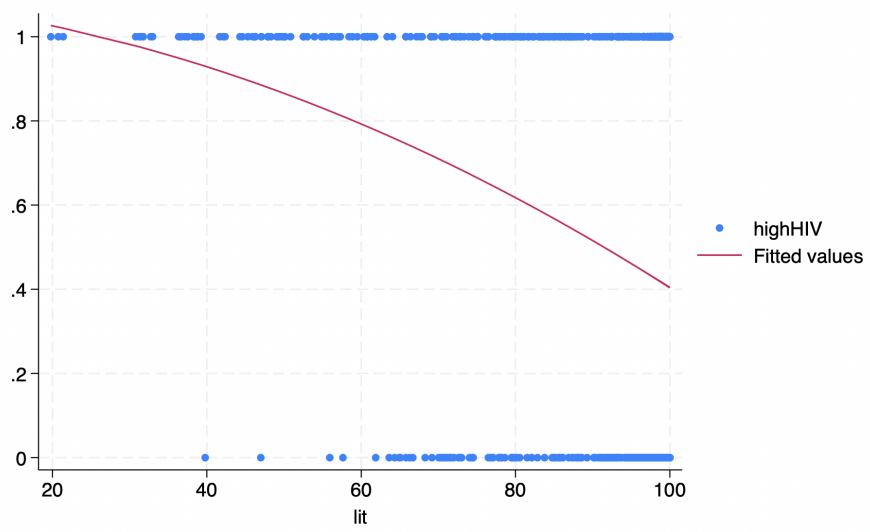


Figure 3. Quadratic model: fitted values of highHIV against the literacy rate.

3.3 Model 3: Probit model

The third model is a probit model estimated by maximum likelihood:

$$\Pr(\text{highHIV} = 1) = \Phi(2.54 - 0.0278\ (\text{literacy rate}))$$

The fitted probabilities are shown in Figure 4. The probit coefficient is not directly interpretable, but it is statistically significant at the 10, 5, and 1 percent levels, indicating that literacy is associated with HIV prevalence. Its negative sign indicates a negative relationship between the literacy rate and HIV prevalence.

To obtain a marginal effect, we compute Y_1, the predicted probability that highHIV equals one when the literacy rate equals its sample mean ($\bar{X} = 84.4$), and Y_2, the predicted probability when the literacy rate equals $\bar{X} + 1$, and then take the difference. $Y_1 = \Phi(2.54 - 0.0278 \times 84.4) = \Phi(0.193) = 0.5080$, and $Y_2 = \Phi(2.54 - 0.0278 \times 85.4) = \Phi(0.165) = 0.5040$. Thus $Y_2 - Y_1 = -0.0040$: evaluated at the mean literacy rate, a one-percentage-point increase in literacy changes the probability of high HIV prevalence by −0.0040. Relative to the baseline probability, $(0.0040 / 0.5080) \times 100 = 0.787$ percent, which we consider economically small.

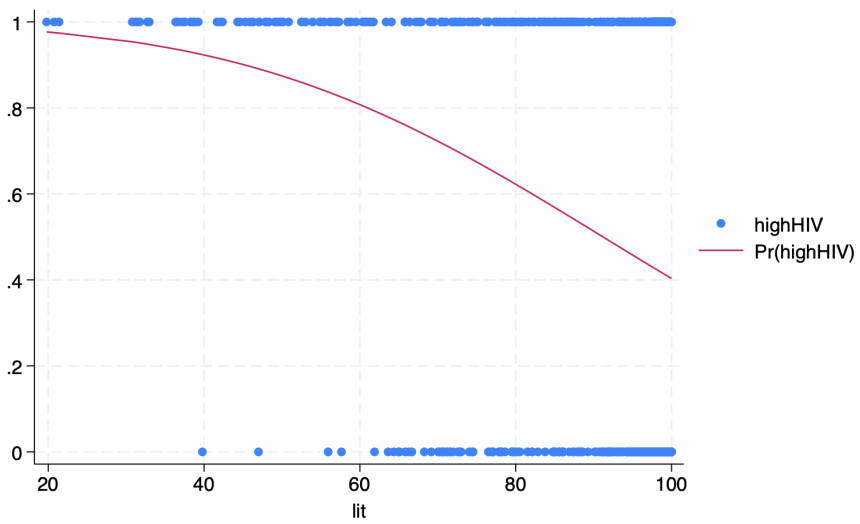


Figure 4. Probit model: predicted probability of high HIV prevalence against the literacy rate.

3.4 Model 4: Fixed effects model with life expectancy control

The final model is a linear fixed effects regression that controls for life expectancy, estimated by OLS with robust standard errors:

$$\text{highHIV}_{it} = 1.708 - 0.0048\ (\text{literacy rate})_{it} - 0.0267\ (\text{life expectancy})_{it} + \beta_3 Z_i + \beta_4 W_t$$

Here Z_i denotes country fixed effects and W_t denotes year fixed effects. We do not present a scatter plot for this model because the combination of panel fixed effects and the additional control makes a two-dimensional plot uninformative.

Holding constant all factors that vary across countries and years, as well as life expectancy, a one-percentage-point increase in the literacy rate decreases by 0.0048 the probability that a population is located in an area where HIV prevalence exceeds 0.4 percent. Similarly, holding constant country and year factors and the literacy rate, a one-year increase in life expectancy decreases that probability by 0.0267. The literacy rate, our variable of interest, is not statistically significant at the 10, 5, or 1 percent level. The life expectancy control is statistically significant at the 10 and 5 percent levels but not at the 1 percent level.

For the literacy rate, $(-0.0048 / 0.4) \times 100 = -1.2$ percent relative to the median HIV prevalence, which is economically small. For life expectancy, $(-0.026 / 0.4) \times 100 = -6.5$ percent relative to the same reference point, which we also consider economically small.

3.5 Summary of results

Table 1. Regression results

Dependent variable: High HIV prevalence (highHIV)	(1) Linear probability model	(2) Polynomial regression	(3) Probit model	(4) Fixed effects
Literacy rate (X_1)	−0.0090703*** (0.0007758)	−0.0090703 (0.00559009)	−0.0278545*** (0.0032969)	−0.0048 (0.0034)
Life expectancy at birth (X_2)				−0.0267** (0.011)
Literacy rate squared (X_3)		−0.0000486 (0.0000434)		
Constant	1.321257 (0.066637)	1.084194 (0.1725155)	2.540273 (0.3020206)	1.708 (0.702)
Country and year fixed effects	No	No	No	Yes
Sample size (n)	761	761	761	761
Adjusted R^2†	0.086	0.086	0.071	0.77

Notes: Standard errors are in parentheses. Models 1, 2, and 4 are estimated by OLS with robust standard errors; Model 3 is estimated by maximum likelihood.

*** Statistically significant at the 10, 5, and 1 percent levels.

** Statistically significant at the 10 and 5 percent levels, but not at the 1 percent level.

† For the probit model (3), the reported value is the pseudo R^2.

4. Discussion

Of the four specifications, the fixed effects model with the life expectancy control best captures the causal effect of literacy on HIV prevalence. It includes the most controls for omitted variable bias, namely country fixed effects, year fixed effects, and life expectancy. Its adjusted R^2 of 0.77 is also substantially higher than that of the other models, indicating a better fit: the adjusted R^2 measures the share of the variation in Y explained by the regressors, adjusted for the loss of degrees of freedom.

5. Threats to Internal Validity

5.1 Omitted variable bias

One variable we do not control for that could generate omitted variable bias is gender, specifically whether a respondent is female (Z = 1) or not (Z = 0). The World Bank reports that young women in developing countries are the population most susceptible to HIV, primarily because some are compelled to turn to sex work as a source of income and because older clients coerce them not to use condoms. Prevalence of HIV (Y) and female gender (Z) are therefore

positively correlated. Literacy (X_1) and female gender (Z) are negatively correlated, as men have historically had greater opportunities to pursue education, especially in developing countries. Because $Cov(X_1, Z) < 0$ and $Cov(Y, Z) > 0$, the omitted variable produces a negative bias. If the true effect of literacy on HIV prevalence is negative, the estimated coefficient on literacy is therefore overstated in magnitude as a result of omitting gender.

5.2 Sample selection

Sample selection arises if the samples used to measure X or Y are not representative of the population of interest. The literacy data come from the UNESCO Institute for Statistics, which collects them annually through national censuses and household surveys and uses the Global Age-Specific Literacy Projection Model (GALP) for countries without recent literacy data. These samples may nonetheless be biased: illiterate individuals are less likely to respond to written surveys, and, because of lower incomes, they are more likely to be unhoused or otherwise missed by censuses and household surveys. This would produce non-response bias in the literacy variable (X_1). Because non-response removes observations with low literacy, it does not bias the coefficient on X_1 in this case, but it reduces the sample size, raises standard errors, and makes predictions less reliable for values of X_1 outside the range of the observed data.

5.3 Simultaneous causality

Simultaneous causality arises if a country's literacy rate affects its HIV prevalence while HIV prevalence simultaneously affects its literacy rate. A higher prevalence of HIV (Y) likely indicates that a country has fewer resources and a less structured school system. In addition, family members may need to care for those affected by HIV and thus have less opportunity to pursue education, which lowers literacy (X_1). At the same time, lower literacy is likely to raise HIV prevalence. Because both channels operate in the same direction, simultaneous causality would cause the coefficient on X_1 to be overestimated by exaggerating the effect of X_1 on Y.

5.4 Measurement error

Measurement error arises if either HIV prevalence or the literacy rate is misreported. The HIV data come from UNAIDS, and the World Bank cautions that estimates of disease prevalence and incidence are often unreliable and incomplete, because national health authorities differ widely in their capacity and willingness to collect or report information (World Bank, n.d.-c). If countries underreport HIV prevalence, or if individuals falsely report being HIV-negative

because testing is inaccessible, the resulting error in Y would be nonrandom. If prevalence were underreported specifically in countries where it is highest, the correlation between X and Y would not be fully captured, the slope of the regression line would appear less steep, and the coefficient on X would be underestimated.

6. External Validity

This study is most relevant to policymakers and officials seeking to reduce HIV rates through health literacy programs, particularly in lower-income and developing countries and especially in Sub-Saharan Africa, which has the highest HIV rates in the world. Our dataset contains sufficient observations on both HIV prevalence and literacy for Sub-Saharan Africa, so we regard the study as having high external validity for countries of similar socioeconomic status.

Zimbabwe, which is in Sub-Saharan Africa, illustrates this point. Zimbabwe has a substantial amount of missing data, so we cannot interpret the causal link between HIV prevalence and literacy for Zimbabwe on its own. However, because the study has external validity in this context, our results can be used to characterize similar populations. Finally, because the data cover 2000 to 2021, the results lack external validity for earlier periods, especially given the medical advances of recent decades that have substantially lengthened the lives of people living with HIV.

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